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5.2.8. Box Plot for Age by Survived

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Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"** .
3. Remove the default subtitle with **plt.suptitle ("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

import pandas as pd

```
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Boxplot of Age grouped by Survived
data.boxplot(column='Age', by='Survived')

# Remove default subtitle
plt.suptitle("")

# Set title and axis labels
plt.title("Age by Survival")
plt.xlabel("Survived")
plt.ylabel("Age")

# Show plot
```

plt.show()

`data = pd.read_csv('Titanic-Dataset.csv')`

Average time
0.239 s
239.00 ms

Maximum time
0.239 s
239.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 239 ms Debug

Expected output

Actual output

Age by Survival

This box plot shows the distribution of age for survivors (Survived=1) and non-survivors (Survived=0). The y-axis represents age from 0 to 80. For non-survivors, the median age is approximately 28, with a box from 22 to 35 and whiskers from 0 to 50. For survivors, the median age is approximately 28, with a box from 22 to 35 and whiskers from 0 to 60. There are several outliers for non-survivors at higher ages (around 60-70).

Age by Survival

This box plot is identical to the expected output, showing the distribution of age for survivors and non-survivors. The y-axis represents age from 0 to 80. For non-survivors, the median age is approximately 28, with a box from 22 to 35 and whiskers from 0 to 50. For survivors, the median age is approximately 28, with a box from 22 to 35 and whiskers from 0 to 60. There are several outliers for non-survivors at higher ages (around 60-70).

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